IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

POWER SYSTEM DYNAMICS STABILITY AND CONTROL

Friday, 15 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

1.

a)

i) What are switching surges in electric power transmission ?

[2]

ii) How are they characterised ?

[2]

iii) How are they different from lightling surges ?

[2]

iv) Why are they so important for network element design such as power circuit breakers?

[2]

v) What kind of surge is most important in transformer protection design?

[2]

b) This part relates to large synchronous machine construction

i) Why are the excitation windings housed on the rotor ?

[2]

ii) Why are the armature windings housed in the stator ?

[2]

iii) Why should the rotor be of solid construction ?

[2]

iv) Why is the stator laminated ?

[2]

v) Why are the stator conductors of hollow tubular structure?

[2]

vi) How is the frequency of stator voltage related to the rotor speed and the number of poles ?

[2]

c) Usually the choice of generation voltage of a large synchronous generator is restricted between 15-25 kV. Explain with reasons to justify this choice.

[3]

[25 marks]

2.

a)

i) Why do the steam turbine driven generators have round rotor?

[2]

ii) Why do the hydro turbine driven generators have salient rotor ?

[2]

iii) Why is it not very useful to equip a hydrogenator with high gain fast acting speed control system without a transient gain reduction mechanism?

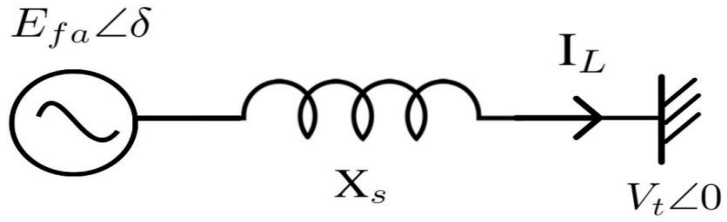
[4]

iv) How can a hydro generator favourably contribute to the rotor angle stability ?

[2]

b)

(i) Starting from the basic circuit equations, establish the power angle relations of a round rotor synchronous generator. You may take the help of the following circuit diagram of a generator connected to network with resistance neglected.



[7]

c)

(i) The above generator windings in star configuration are connected to the grid at 23 kV (line to line) voltage (*Vt*). The rated power is 500 MW, power factor is 0.85 (lagging). The synchronous reactance (Xs) is 1.8 pu on its rated base of 588 MVA. Find out, at full output power:

ii) the excitation phase voltage in kV

[2]

iii) load angle (δ) in degrees

[2]

iv) load angle stability margin in degrees

[2]

v) total terminal reactive power in MVAR

[2]

[25 marks]

3.

a) This part refers to synchronous machine:

i) What is a rotating reference frame in the context of synchronous machine modelling and analysis ?

[2]

ii) What are the advantages of rotating reference frame over its static counterpart when analysing behaviour of stator-rotor magnetic interactions ?

[3]

iii) Why is one rotating reference frame not enough in the context of multimachine stability study ?

[2]

iv) What are the stator transients in the context of synchronous machine operation?

[2]

v) How are the stator transients represented for stability studies in rotating reference frame?

[2]

vi) Why are the stator/network transients traditionally neglected in first swing stability study for a large synchronous machine ?

[3]

b) This part refers to asynchronous machine:

i) Why can not these stator transients be neglected in converter interfaced wind farm stability study?

[3]

ii) Why is Type 3 (DFIG) cheaper than Type 4 (PMSM) for the same rated output capacity ?

[3]

iii) Why is angle stability not as concerning as oscillatory or synchronisation stability in the operational context of grid connected wind farm ?

[3]

iv) How is the wind *wake* represented for fast frequency response provision ?

[2]

[25 marks]

4.

a)

i) Starting with the definitions, narrate how capacity factor in a wind turbine differs from power conversion coefficient (Cp) ?

[4]

ii) Why does an offshore wind turbine of similar installed power capacity produce more energy over a year over its onshore counterpart?

[2]

iii) What is the tip-speed ratio in the context of wind energy conversion and control ?

[2]

iv) Why is it so important to have the optimal tip-speed ratio in wind turbine operation ?

[2]

v) When is the tip-speed ratio control not effective ?

[2]

b)

i) What are the fundamental differences between the pitch and stall control in wind farm operation ?

[4]

ii) How is the pitch control for WTGs modelled for stability studies ? You may take help of suitable block diagram to answer this question.

[3]

iii) How are the stall control mechanisms incorporated in modern wind turbine fleets?

[2]

iv) How is the rotor flux of a permananent magnet synchronous machine (PMSM) type wind turbine modelled for stability studies ?

[2]

v) How are the turbine and generator shafts of a DFIG (Type 3) modelled for stability studies?

[2]

[25 marks]